AI
COMPUTING

Breaking the Data Center Walls: Unifying AI Fabrics for Agentic AI at the Edge

Prakash Jain

Director, Architecture & Design Engineering (AI Infrastructure)

Celestica

OCP PLATINUM Member

Agenda

Breaking the “Data Center Walls”…

01 **The Agentic Shift**
Transitioning from centralized, stateless inference to persistent agentic workflows distributed at edge.

02 **Network Fabric Requirements**
Meeting stringent metrics of high bandwidth, ultra-low latency, and lossless RDMA reliability, even across WAN/RAN.

03 **Proposed Solution & Deep Dive**
Unifying AI fabrics for optimized memory access across domains for Agentic AI.

04 **Future Work & Call to Action**
Further research work and call to action.

Goal: Expand the AI Fabric beyond Data Center Walls to enable next-gen Agentic AI at the edge.

The Agentic Shift

AI is moving from stateless, centralized inference to persistent, distributed agentic workflows.

Traditional Stateless Inference:

- **Transient Execution:** One-and-done request/response cycles.
- **Stateless Context:** No native memory retention across steps.
- **Centralized Topology:** Bound to isolated, massive data centers.

The Agentic AI:

- **Persistent Workflows:** Multi-turn, self-correcting agent chains.
- **Stateful Execution:** Continuously updated state tracking with memory retention.
- **Distributed at Edge:** Executed closer to environments and users.

The Problem: AI Compute Infrastructure is trapped within the **"Data Center Walls”,** so it doesn’t scale for Distributed Agentic AI

The Scaling Dilemma for Agentic AI Compute

Strict Infrastructure Requirements for Distributed Agentic AI Workflows

Persistent State

The need for 4-tier Persistent **Distributed KV Caches** (PD-KVC) to sustain context across sessions.

Fault Tolerance

Transitioning from traditional heavy DC checkpointing to lightweight, **continuous Agent State Journaling** (ASJ).

Reliable Agent Coordination Channel (ACC) acting as the vital infrastructure "glue" for planning and executing agents.

Network Fabric

AI Compute needs **High Performance Network** with strict metrics requirements:

Bandwidth ≥ 400–800 Gbps per port.

Latency < 1-100 µs (target ~250 ns).

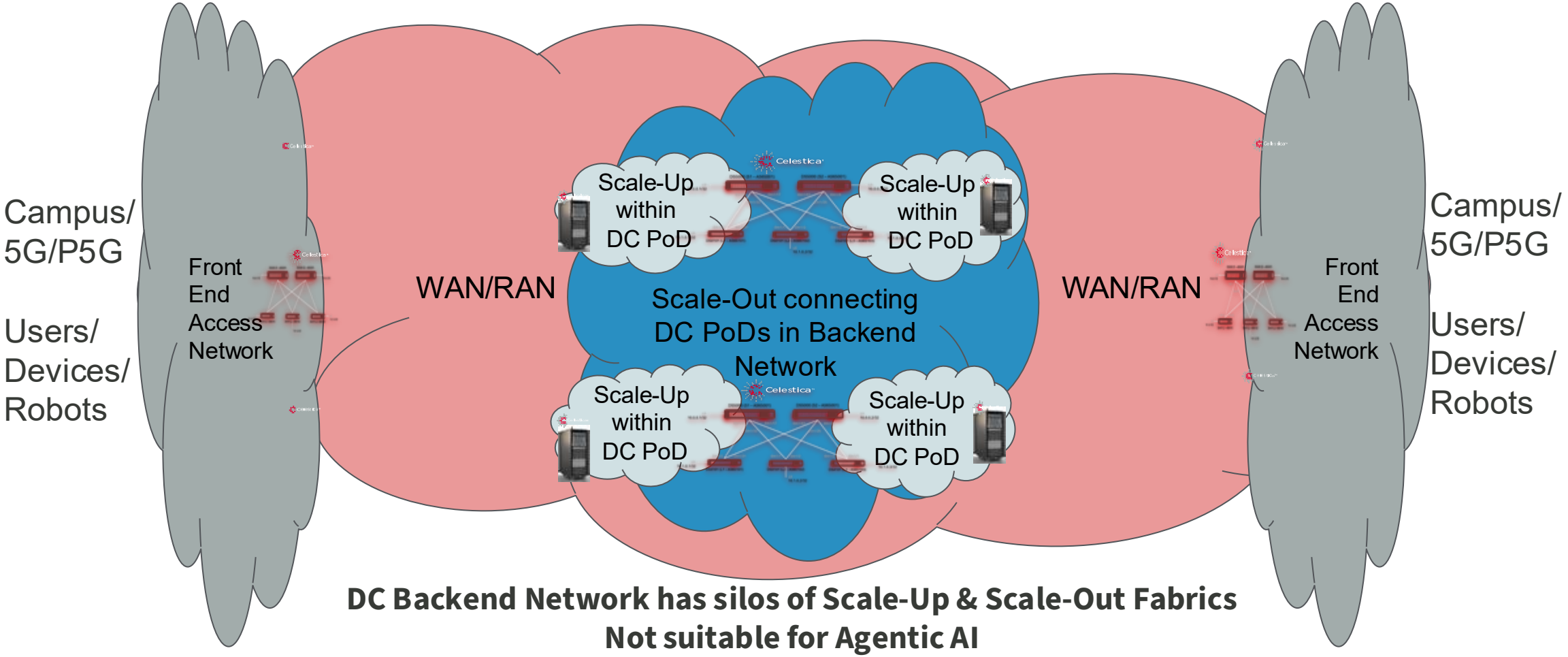
Reliability Lossless (RDMA preservation).

Consistency Deterministic, jitter-free performance under load.

AI Compute = Compute + Network (supporting In Network Collectives Computation)
With stringent network performance metrics (even over WAN/RAN) to support Distributed Compute & Memory

Network Bottleneck for Agentic AI Compute

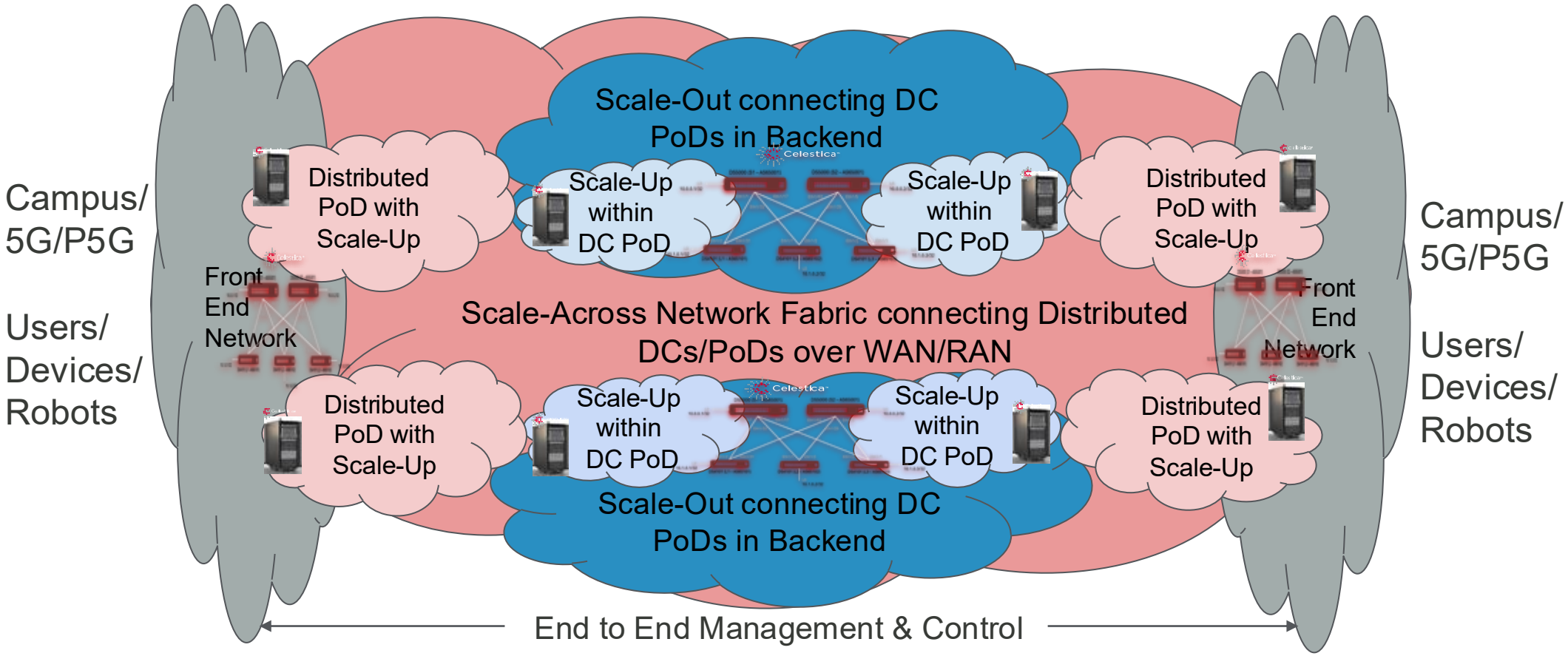
AI compute is in far Hyperscale DCs, access via WAN/RAN does not meet the required performance



DC Backend Network has silos of Scale-Up & Scale-Out Fabrics
Not suitable for Agentic AI

Need Distributed AI Compute over Backend Network

Distributed PoDs needed to extend towards Edge over WAN/RAN for AI Inference (Beyond the Data Center Walls)



Proposed Solution: Unified "Scale-Across" Fabric

The New Paradigm

Unifying heterogeneous AI fabrics

Integrating Scale-Up, Scale-Out & Scale-Across architectures into a single cohesive fabric connecting geographically diverse sites.

Pillar 01

Distributed Racks & PoDs

Enabling **dis-aggregated compute** architectures at edge to operate seamlessly as unified resource pools.

Pillar 02

Agentic Intelligence

Integrating closed-loop telemetry to automatically manage, **monitor, and optimize cross-fabric** Accelerators endpoints.

Pillar 03

End-to-End Control Plane

Integrated transport and orchestration capabilities that **tie multi-domain networking elements** together.

Pillar 04

Inter-Domain Coordination

Unifying protocols across multiple domains with varied local/regional policies, cloud layers, and physical layer technology barriers.

Distributed Racks & PoDs

Enabling dis-aggregated compute architectures to interconnect seamlessly as unified resource pools.

COMPONENT 01

Dis-aggregated Compute

Breaking down traditional server boundaries to **pool processing power, memory, and storage** independently across the entire fabric.



COMPONENT 02

Unified Resource Pools

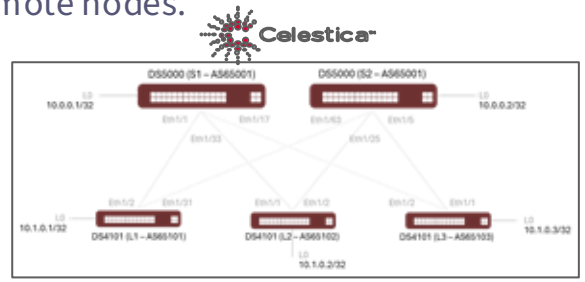
Dynamic orchestration layers that treat geographically diverse physical hardware as a single, cohesive, and fluidly allocatable logical system.



COMPONENT 03

Seamless Network Interconnect

High-bandwidth, ultra-low latency optical and physical backplanes **linking distributed racks without bottlenecking** remote nodes.



Agentic Intelligence

Integrating intelligent closed-loop telemetry to automatically manage, monitor, and optimize cross-fabric Accelerators Endpoints.

CAPABILITY 01

Closed-Loop Telemetry

Real-time telemetry-driven orchestration that **monitors active agent** sessions and proactively optimizes system performance across domains.

CAPABILITY 02

Autonomous Workflow Actions

Distributed hierarchical agents implementing closed-loop control loops to execute complex **multi-step reasoning workflows** across domains.

CAPABILITY 03

Cross-Fabric Optimization

End-to-end trace validation and optimization across distributed compute at edge locations, ensuring logical state coherence and **consistent secured policies across the domains**.

End to End Unified Fabric Control Plane

To unify heterogeneous backend fabrics in one global fabric

Unified Control Plane is needed to expand Infrastructure beyond "Data Center Walls."

Control Plane needs to decouple Endpoint Identifiers (AccIDs, Non-IP/Mac or IP/Mac) from Network Routing Mechanisms (IP/Mac Based).

Requirement 01

Accelerators Addressing

Supports both IP/Mac and non-IP/Mac based addressing for accelerators (AccIDs).

Requirement 02

Workload Mobility

Enables seamless workload mobility across heterogeneous fabrics for operations and reliability.

Requirement 03

Multi-Tenant Isolation

Uses Instance-IDs (IIDs) to support multi-tenants, isolation and INC integration across multiple domains.

Requirement 04

Unified Scaling

Allows Scale-Up, Scale-Out, and Scale-Across domains to operate under one framework.

Inter Domain Coordination via Control Protocols

Control Plane Protocol handles network path establishment, changes, avoids congestion and provides reliability for inter domain coordination end to end

Protocol Requirement 01

Registration/Subscription

NICs/Switches (ToRS/RLOCs) register/deregister Accelerators End Points (EIDs) with the control plane when they are available/unavailable.

Remote switches/NICs subscribe/unsubscribe to the control plane for Accelerators End Points (EIDs) with which they need to connect.

Protocol Requirement 02

Notifications/Publications

Real-time notifications or publications for path establishments as new Accelerator joins.

Real-time notifications or publications for path changes (failure/congestion states).

Protocol Requirement 03

Fast Convergence & Resiliency

Sub-microsecond convergence when moving workloads across distributed PoDs.

Multipath and redundancy to switch to alternate paths/compute resources, based on the availability.

Technical Deep-Dive: LISP Protocol Based AI Infra

Example of LISP based Control Plane AI Infra to scale across domains with flexibility & secured isolation

Domain 01

Scale-Up (eg Rail optimize)

- Scale-Up NIC/switch registers the mapping of each unique Acc-Id (non-IP EID) with NIC/Switch's IP (RLOC)
- Control Plane maps non-IP AccIDs to Scale-Up switches (RLOCs), publishes updates.
- Switch/NIC subscribes and facilitates intra-PoD traffic supporting up to 1,024 accelerators in a PoD to sustain sub-microsecond latency needed for load/store.

Domain 02

Scale-Out (eg Leaf-Spine)

- Scale-Out NIC/switch registers the mapping of each unique Acc IP/MAC (EID) with NIC/Switch's IP (RLOC)
- Control Plane maps Acc IP/MAC Address to Scale-Out Switches IP (RLOCs), publishes updates
- Switch/NIC subscribes and interconnects PoDs within a site to support high-speed clustering using IP encapsulated traffic for RDMA

Domain 03

Scale-Across (Multi DCs)

- Scale-Across GW Switch registers the mapping of each site's subnet address (EID) with GW Switch's IP Address (RLOC).
- Control Plane maps site subnets to Scale-Out Switches IP (RLOCs), publishes updates
- GW Switch subscribes and interconnects distributed DCs over WAN/RAN using IP encapsulation to unify global AI Fabric infrastructure.

Convergence & Segmentation

Logical Segmentation

- LISP decouples location from identity, provides pull-based protocol with Pub-Sub based dynamic updates within the logical segments/instances.
- Leverages Instance-IDs (IIDs) to guarantee strict tenant-level isolation.
- Control plane maps Collective Computation (INC) domains to logical segments to offload collectives operations to network.

Data Plane

Encap & Tunneling

- LISP supports any data plane encapsulation tailored to transport requirements.
- Control plane maps Instance-IDs to data plane Virtual networks to allow data plane encapsulations & tunnelling.
- Control plane maps Instance-IDs (IIDs) directly to physical multicast groups (G-RLOCs) to support underlay multicast.

- Control Plane Protocols (e.g. LISP) provide the scalable, unified fabric control plane needed to converge heterogeneous AI Fabrics.
- The End to End Control Plane unifies distinct scaling domains (Scale-Up, Scale-Out, Scale-Across with flexibility of choosing any combinations of scaling domains) into one cohesive AI network fabric and interdomain coordination.
- Control Plane Services provides coordinated high-performance for AI training and inference.

Technical Deep-Dive : Scale-Across Architecture

Distributed Scale-Across Architecture unifying Heterogeneous Fabrics with End to End Control & Management Plane

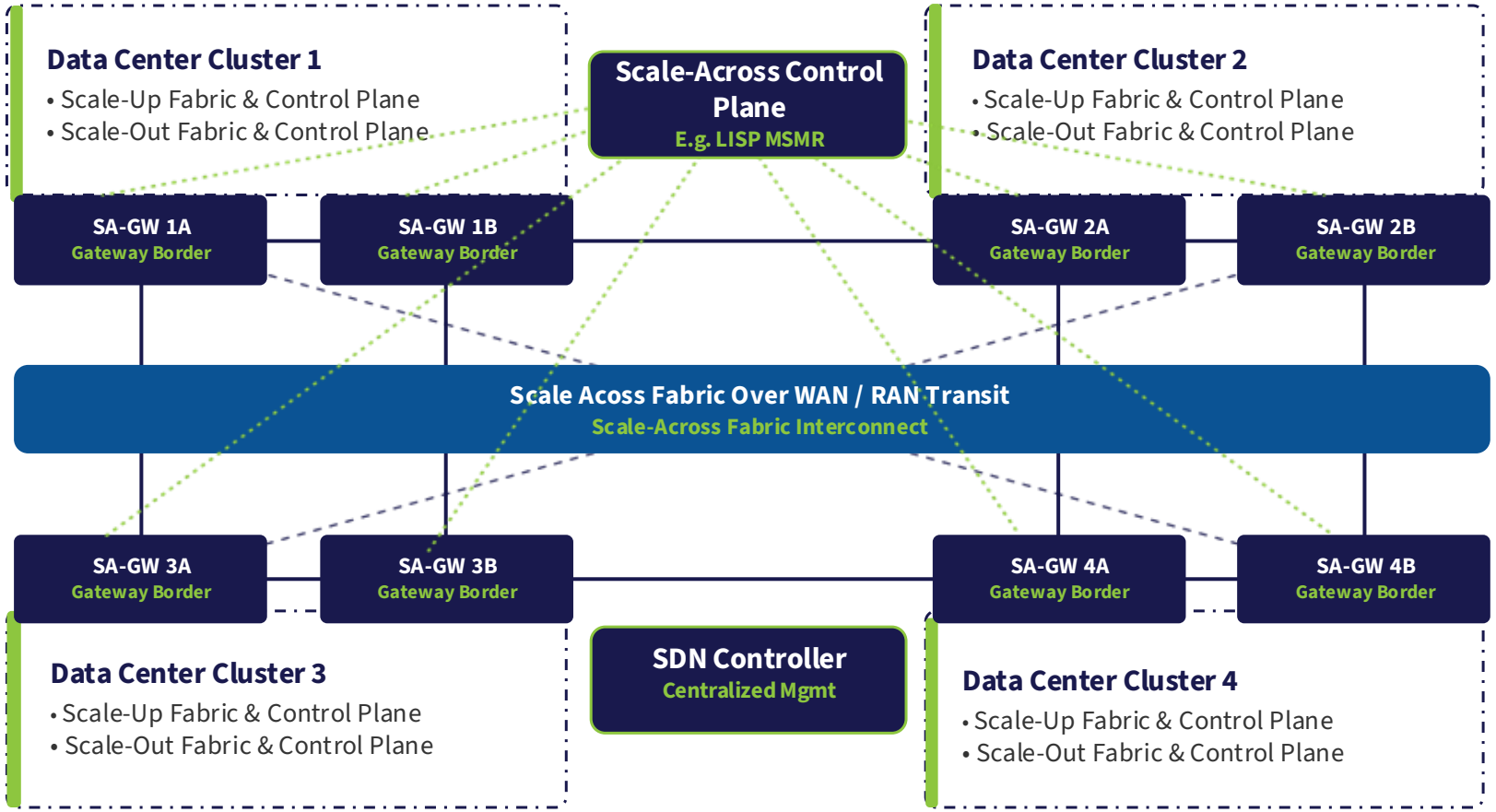
System Overview

Scale-Across Fabric

A multi-DC hybrid architecture interconnecting clusters over a high-throughput WAN/RAN transit mesh.

Distributed Compute:
To enable seamless distributed AI execution, distributed PoDs orchestrate dynamically across high-throughput clusters. Each cluster leverages Scale-Up compute engines linked to Scale-Out routing switches, unified by resilient Control Plane. Fully meshed Scale-Across Gateway Borders provides connectivity to WAN/RAN transit network.

Unified Control Plane & SDN Controller:
Provides globally synchronized control & management to manage Unified Distributed Fabric seamlessly across WAN/RAN.



Technical Deep-Dive: Control Plane Protocols Options

Evaluating SRv6, BGP-EVPN, and LISP protocols for low-latency, high-bandwidth AI network fabrics

Segment Routing v6 (SRv6)

IPv6 based Source-Routing Paradigm: Encodes deterministic path steering & network-level service chains directly into stacked IPv6 packet headers.

Backend Network Accelerators Scaling Fit

- Requires native IPv6 support globally across the transport fabric.
- Significant header stack overhead limits effective hardware throughput.
- Source-baked decisions degrade real-time traffic engineering agility.

BGP-EVPN

Push-Based Control: Widely deployed standard that preemptively floods routing state to all endpoints across the fabric.

Backend Network Accelerators Scaling Fit

- State distribution is too heavy-weight for high-frequency dynamic paths.
- Scale issues stem from maintaining complete prefix tables on every node.
- Lacks rapid sub-microsecond failover for ultra-sensitive collective operations.

LISP (Locator/ID Separation)

Pull-Based Dynamic Mapping: Decouples Endpoint IDs from Routing Locators, resolving paths on-demand via a mapping database.

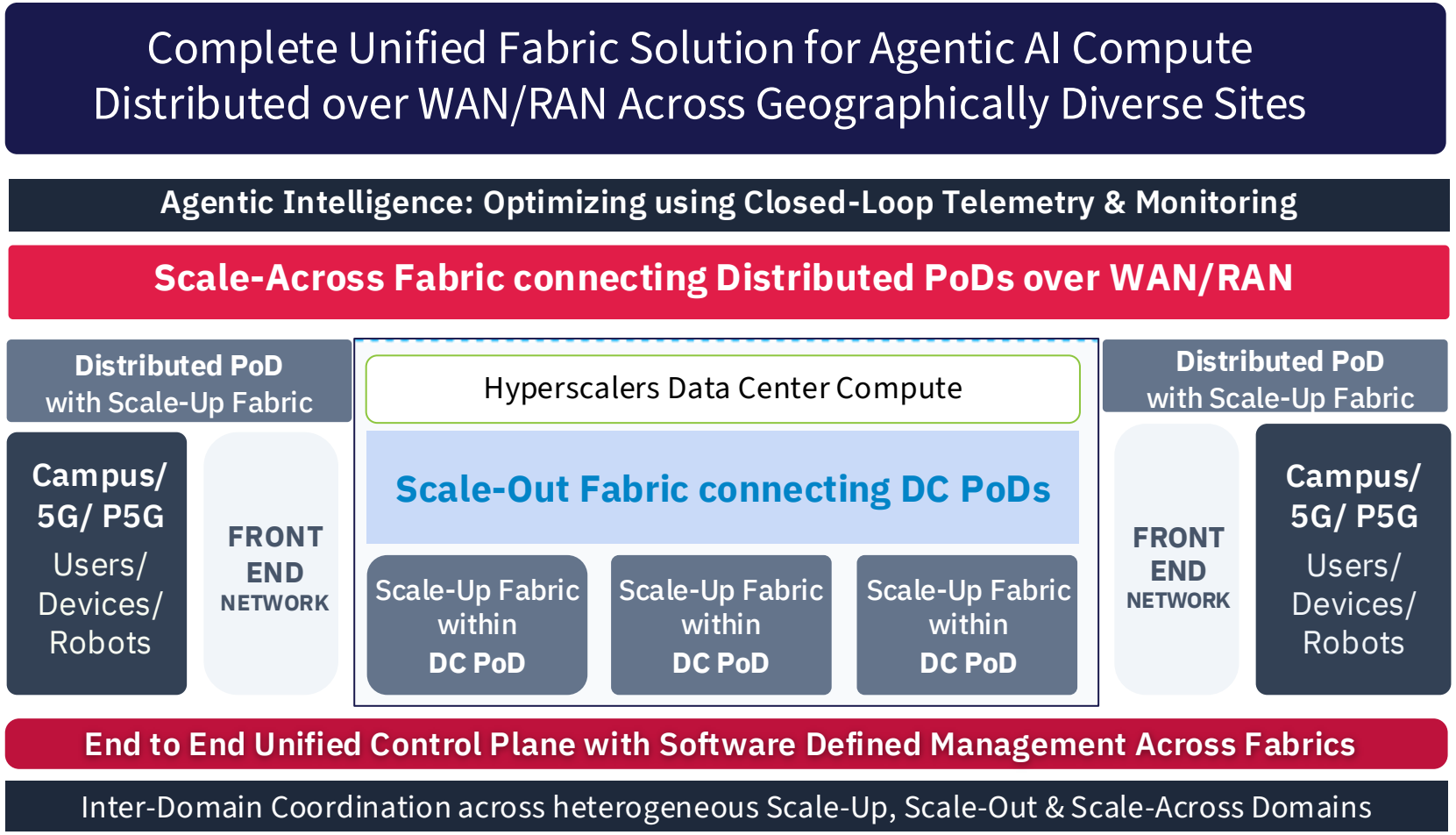
Backend Network Accelerators Scaling Fit

- **Extremely lightweight** control-plane state on nodes; on-demand, no global flooding.
- Seamlessly supports non-IP/MAC and IP/MAC based Accelerator IDs over generic IP underlays.
- Built-in pub/sub architecture guarantees sub-microsecond convergence.

Recommendations for Backend Network for Accelerators Connectivity:

- **Choose LISP** if workload requires ultra-low latency, dynamic scale-across coordination, and lightweight pull-based resolution to bypass the processing limits of classic push based protocols tables.
- **Choose SRv6 / BGP-EVPN** if you are building standardized multi-tenant virtualization fabrics and have infrastructure natively optimized to process high-overhead protocols and push-based tables/resources.

Recap: The "Scale-Across" Solution



Pillar 01 · Distributed PoDs & Fabric
Enabling dis-aggregated compute architectures to operate seamlessly as unified resource pools.

Pillar 02 · Agentic Intelligence
Integrating closed-loop telemetry to automatically manage, monitor, and optimize cross-fabric endpoints.

Pillar 03 · End-to-End Control Plane
Integrated transport and orchestration capabilities that tie multi-domain networking elements together.

Pillar 04 · Inter-Domain Coordination
Unifying operations across varied regional policies, cloud layers, and technology barriers.

Future Research & Evolution

Need to build open, low power, more efficient physical layer hardware for next-gen performance, supported by Open Control Plane Software.

RESEARCH 01

Silicon Photonics

CPO/ELS architectures reducing power (<3pJ/b).

RESEARCH 02

3D Photonic-ASIC Integration

Moving optical engines under host chips (100x bandwidth gains).

RESEARCH 03

Precision Timing Intelligence

Picosecond-level timing (6G RAN) for global synchronization with unified control & management.

Call to Action

Agentic AI requires transition from isolated DC AI Compute Fabrics to Globally Coordinated Unified Fabric of Distributed AI Compute.

ACTION 01

Standardize HW/SW/Protocols

Standardize "Scale-Across" & “Unified AI fabric” frameworks via collaboration across Open forums like **OCP ESUN, UEC, SONiC, IETF**.

UE-Specification-1.0.1.pdf

OCP ESUN - Network Operator Requirements - Base Specification Rev 1.0

ACTION 02

Adopt Unified Fabric Control Plane

Adopt the **open unified AI fabric, control plane and protocols** for distributed Agentic AI.

draft-jain-lisp-network-ai-infra-04 - LISP-Based Network for AI Infrastructure

ACTION 03

Support Anti-Fragmentation

Join the OCP effort to prevent industry fragmentation of AI Infrastructure via collaboration across Industry & Academia in groups such as **AICC & AI HW-SW-CoDesign**.

AI Fabrics Perspective and Modeling

Q&A

Thank You!

Questions/Comments?

Acknowledgement: Appreciate all cross-industry collaboration (OCP, IETF, UEC, SONiC).


Thank
You!
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